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An Auditable Global Research Agenda for Social-Media and Generative-AI Restrictions Affecting Youth

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Abstract

Governments and schools are rapidly restricting phones, social-media accounts, and generative-AI tools for young people, and rules for AI companions are now emerging at the model, feature, and account level. Yet the word *ban* collapses interventions that act on different objects, settings, ages, and mechanisms. This makes policies across the global landscape difficult to compare and encourages a basic inferential error: conflating announced rules with delivered treatments. In this position paper, we review a subset of related work published through August 2026. Several recent studies report that some restrictions reduce the targeted access or use, which, in some cases, has translated to positive downstream effects on learning, well-being, discipline, and safety. In other cases, these downstream reports are mixed, time-varying, or not yet identifiable. To evaluate the effectiveness of age-based technology-restriction policies, we propose TRACE, a minimum evaluation standard covering **T**reatment, **R**each, **A**daptation, **C**onsequences, and **E**quity/ethics/exit, with auditable youth participation.

1. What Defines a "Ban"?

Communities ranging in scale from small classrooms to large countries have implemented policies that aim to limit young people's use of technology in some way. UNESCO reports that, as of March 2026, 58% of countries (served by 114 education systems), have national school-phone restrictions, although the details vary substantially across these contexts (UNESCO Global Education Monitoring Report, 2026). This statistic reflects a rapid increase in the implementation of restrictions, up from 24% of countries in June 2023. Other policies seek to limit young people's technology use outside of school, with Australia, most notably, requiring covered social-media platforms to take reasonable steps to prevent people under the age of 16 from holding accounts (Australian eSafety Commissioner, 2026). School systems are also moving from blocking all use of generative AI toward a heterogeneous mix of regulations and sanctioned experimentation (Liang et al., 2026). Norway, for example, now recommends that pupils in grades 1–7 generally not use AI in school, while allowing cautious, teacher-supported introduction at later stages (Norwegian Ministry of Education and Research, 2026).

This collection of policies reflects a wide range of motivations, values, and approaches. A phone-away-during-class rule changes access in a bounded setting; a platform age-gate changes account access across the day; a school-network block leaves personal devices and home use available; and a ban on submitting AI-generated work targets a use-case, not a technology. Beyond access restrictions, product redesign can shift responsibility from individual users to platforms by changing defaults or limiting risky features. Thus, understanding this policy landscape entails examining restrictions that cannot all be grouped under the umbrella term *ban*. At minimum, a global map must distinguish between *access restrictions* and *design restrictions*:

- **Access restrictions**: aim to limit or completely remove technology use from the day-to-day activities of youth, which can be enforced through mechanisms such as physical device confiscation, network blocks, age-tiered access, and more.
- **Design restrictions**: mandate product design changes which aim to make young people's interactions with technology less harmful, for example, by adjusting default settings or limiting features that promote compulsive use.

2. What Defines a Ban's Success?

The success of any age-based technology restriction should be measured according to its downstream impacts on young people, including side effects and unintended consequences (if any). The core position of our paper is that:

> A restriction should not be called effective merely because it exists, is enforced, or changes access. It succeeds only relative to an explicit objective and counterfactual, after accounting for implementation, substitution, unintended effects, distributional impacts, and time.

This position is deliberately neither *pro-ban* nor *anti-ban*. It is pro-identification: policy-makers should be able to tell what treatment children actually received and which outcomes followed because of it.

3. What is Known about the Success of Existing Bans?

Social-media and generative-AI restrictions are being adopted amid an unresolved scientific dispute about whether, and how much, digital technology harms young people. Haidt (2024) argues that the phone-based rewiring of childhood is a major driver of declining adolescent mental health, while Twenge et al. (2020) similarly contend that digital-media harms have been underestimated. By contrast, Odgers (2024) argues that the causal evidence does not support a claim of that magnitude, and other studies report small average associations between digital-technology use and adolescent well-being (Orben and Przybylski, 2019, 2020). Related work emphasizes that content, design, and developmental mechanisms may matter more than undifferentiated exposure (Orben et al., 2024). Policy-facing reviews likewise diverge (Campbell et al., 2024; Rahali et al., 2024). Rather than seek to resolve this dispute, we treat benefits, harms, implementation, adaptation, and distributional effects as empirical quantities that evaluations should identify.

Table 1: Illustrative studies and what they identify.

Setting and design	Verified first stage	Downstream result and interpretive limit
U.S. lockable phone pouches; national quasi-experimental study (Allcott et al., 2026)	Substantial reduction in school phone use	Average test-score effect near zero. Disciplinary incidents initially rose and well-being initially fell before later improvement.
Florida district phone bans; difference-in-differences (Figlio and Özek, 2025)	Presumed reduction in school phone use, not explicitly captured	Suspensions rose initially, especially for African American students, then faded; second-year test scores improved and unexcused absences fell.
Rio de Janeiro classroom-only, then full-day, phone bans; matched difference-in-differences across 325 schools (Lichand et al., 2026)	Schools reporting "most/all" students using phones fell from 52% to 19% (vs. 40% to 15% in comparison schools)	Test scores rose 0.06 SD relative to controls after the full-day phase; the classroom-only phase showed no significant effect. Gains concentrated where use actually fell.
Australia under-16 social-media minimum age; national baseline (Nov. to Dec. 2025) and 3-month panel (Australian eSafety Commissioner, 2026)	Account ownership fell from 52.4% to 42.1%; use of the platforms fell only from 85.9% to 81.5%	Just over half of retained accounts faced no platform age check; daily messaging use rose 11.8%; parents unaware of their child's use rose from 23.3% to 33.3%. Most reported little or no well-being change.
High-school mathematics; randomized access to standard or safeguarded GPT (Bastani et al., 2025)	Supported practice performance rose with both tutors (+48% standard; +127% safeguarded)	Unsupported exam performance fell 17% with the standard tutor; teacher-designed guardrails eliminated the harm.

The relevant evidence base contains two different literatures. *Policy-effect studies* estimate consequences of a restriction. *Technology-effect studies* estimate associations or effects of exposure to a device, platform, or AI system. The latter can motivate intervention but cannot establish that a particular ban works. A 2024 scoping review of 22 school-phone studies across 12 countries found heterogeneous definitions and outcomes, no randomized controlled trials, and little conclusive support for one-size-fits-all bans (Campbell et al., 2024). Newer quasi-experimental studies provide stronger causal evidence, but still require researchers to distinguish between policy adoption, implementation, changes in use, and downstream outcomes. The studies in Table 1 were selected to showcase the diversity of studies in this space and the outcomes they have reported, including positive, null, negative, and dynamic effects across many different types of restrictions.The purpose of showing this range is not to estimate an average effect, but to show why comparisons across contexts are difficult and to motivate a map of the policy landscape.

Four conclusions emerge from Table 1. First, restrictions can reduce targeted behavior without eliminating it: Australia's policy reduced under-16 account ownership, but platform use remained above 80% (Australian eSafety Commissioner, 2026). Second, downstream

effects can be mixed and time-varying: lockable phone pouches initially worsened discipline and well-being before later improvement (Allcott et al., 2026). Third, users may adapt: in Australia, messaging use and parental unawareness increased (Australian eSafety Commissioner, 2026). Fourth, some policies improve targeted outcomes. The studies report later improvements in well-being and modest test-score gains among high-school students, improved scores and attendance in Florida, and a 0.06 SD test-score increase in Rio de Janeiro (Allcott et al., 2026; Figlio and Özek, 2025; Lichand et al., 2026).

The evidence is also geographically and institutionally uneven. National counts tell us where rules exist, not how individual districts and schools interpret them. English-language studies over-represent higher-income settings; school-level enforcement data, vendor logs, age-assurance error rates, disability accommodations, and disciplinary disparities are rarely linked. Appendix A.1 therefore begins the proposed registry with illustrative dated policy records spanning multiple jurisdictions and levels of governance. For generative AI, current work maps rapid policy evolution (Liang et al., 2026) and tests particular learning designs (Bastani et al., 2025), but direct causal evaluations of school AI bans remain exceptionally sparse and AI restrictions raise evaluation problems that access-and-minutes framings cannot capture.

4. TRACE: A Framework for Describing and Evaluating Technology-Restriction Policies

We propose TRACE as a minimum common record for policy description and evaluation. Its central aim is to a policy's theory of change explicit and auditable by articulating the steps it hopes to achieve from the policy text itself through to its eventual downstream outcomes. Rather than treating a policy as a single, self-contained document, TRACE outlines the full chain of implementation, exposure, adaptation, and mechanism, so that each step in this pathway and any causal relationships between these steps can be inspected and challenged. By articulating a policy's theory of change, TRACE supports more rigorous evaluation and clearer communication across the policy lifecycle. TRACE comprises five elements:

T—Treatment specification. Record the restricted object (device, network, account, model, feature, or use); jurisdiction and decision level; ages/grades; time and place scope; start and revision dates; legal force and operative status (enacted, scheduled, in force, enjoined, repealed); exceptions; approved alternatives; and stated mechanism. Preserve the exact policy text. A district that asks phones to remain silent in backpacks and a school that requires all-day lockable pouches are different treatment approaches.

R—Reach and real implementation. Measure whether the rule reaches children in practice: awareness, school and teacher fidelity, platform compliance, availability of personal devices, age-assurance performance, exceptions granted, and enforcement labor. Include a behavioral first stage rather than inferring exposure from policy adoption. Reach can sometimes be measured while minimizing the collection of youth's sensitive data:: consented screen capture with on-device feature extraction and aggregate-only release (Reeves et al., 2020), voluntary donation of platform data-download packages processed locally (Boeschoten et al., 2022), platform transparency reporting, and independent age-assurance audits that report error, exclusion, and circumvention rates by subgroup (Australian Government, 2025). These methods still require assent and consent procedures, data minimization, access and retention controls, distress and disclosure protocols, and safeguards for reportable or illegal

material. Age assurance must be weighed against privacy, accessibility, and security costs rather than optimized for blocking alone.

A—Adaptation and substitution. Pre-specify likely workarounds and replacement behaviors across services, devices, locations, and time. Measure displacement to messaging, gaming, smaller platforms, non-school accounts, VPNs, shared devices, other AI models, or unmonitored use; East Asian gaming curfews, for instance, were met with borrowed adult identities and account-rental markets (Yang et al., 2023). Ask whether the policy changes adult visibility and young people's help-seeking, not only total minutes online.

C—Consequences and causal identification. Link each stated mechanism to proximal outcomes (attention, classroom interruption, assignment process, exposure to unsafe interactions) and distal outcomes (learning, well-being, belonging, safety, attendance, discipline). Use a credible counterfactual and repeated follow-up. For example, Shi and Villarroel (2026) compare the same Chilean students across subjects with and without teacher-imposed phone restrictions and track teachers whose policies changed across cohorts, thereby accounting for stable student and teacher differences. Staggered rollouts, interrupted time series, difference-in-differences, and randomized delayed implementation can be informative when assumptions and spillovers are tested. Report nulls and adverse effects, not only intended outcomes.

E—Equity, ethics, and exit. Disaggregate enforcement and outcomes by age, gender, disability, race/ethnicity, socioeconomic position, language, and prior risk where lawful and safe. Audit privacy burdens, age-assurance errors, accommodations, and exclusionary discipline. Include children in defining benefits and harms, and make that participation auditable rather than declarative. The Lundy model requires space, voice, audience, and *influence* (Lundy, 2007); General Comment No. 25 shows the standard is attainable at scale, with 709 children aged 9 to 22 across 27 countries consulted before drafting and their input mapped to the final text (UN Committee on the Rights of the Child, 2021; Livingstone et al., 2024). Australia illustrates the converse: a parliamentary committee recommended co-designing features affecting young people, and the adopted law proceeded to blanket prohibition regardless (Fraser and Griffiths, 2024). Evaluations should publish influence statements recording which youth recommendations were accepted, modified, or rejected, and why. Exit, finally, is realizable rather than hypothetical: South Korea's 2011 nighttime gaming shutdown was repealed effective in 2022, a complete, dated policy life-cycle whose end is itself an analyzable second shock (Republic of Korea, 2021). Every policy should state review dates and retain/revise/sunset thresholds.

5. Conclusion and future work

Understanding the effectiveness of age-based technology-restrictions requires a shared way to specify restrictions, verify delivery, detect adaptation, estimate consequences, involve the young people affected, and revise policies when evidence changes. TRACE provides that minimum standard, and Appendix A.1 shows that the registry it requires is buildable today. AI companions are an immediate test case: recent laws, proposals, regulatory inquiries, and provider rules target different policy units and mechanisms (California State Legislature, 2025; United States Congress, 2025; Federal Trade Commission, 2025; Character.AI, 2025), while their effects depend on interaction design as well as access (Bastani et al., 2025). Future work should apply TRACE prospectively to measure reach, circumvention, interaction

harms, model changes, substitution, and exit criteria. Together, a living global registry and prospective evaluations can replace policy-label debates with child-centered evidence about what works, for whom, through which mechanism, and at what cost.

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Appendix A. Appendix

This appendix translates TRACE into two practical tools for cumulative research. Appendix A.1 presents an illustrative, rather than exhaustive, registry spanning restrictions on devices, accounts, networks, models, features, and uses. It distinguishes announcement and enactment from implementation and exposure, links policies to available evidence, and records adaptation and legal change. Appendix A.2 provides a question bank for applying the same standard prospectively across policies and settings.

A.1. An initial registry of youth-directed technology restrictions

Statuses are as of August 2026 and were compiled from the cited primary legal texts and evaluations. Legal statuses in this area change monthly, so production registry records should carry `valid_from`, `valid_to`, `status_as_of`, and archived-source fields. "No evaluation found" means that no policy-specific outcome evaluation was identified in this illustrative review, not that no such evaluation exists or that the policy has no effect. "First stage" means a measured change in the restricted behavior, not legal enactment. The final column includes both reported observations and prospective implementation considerations; a production registry should separate these using `evidence_status` values such as `observed`, `reported`, and `hypothesized`, together with an `adaptation_source` field.

Policy; instrument	**Object; ages; scope; stated mechanism**	**Dates; status (Aug. 2026)**	**Evaluation / measured first stage**	**Adaptation notes**
S. Korea: Youth Protection Act art. 26 ("shutdown law"); national statute (Republic of Korea, 2021)	Online-game access; under 16; 00:00 to 06:00 daily; provider-side blocking to protect sleep	Eff. Nov 2011; parental-choice option 2014; repealed 2021, ceased Jan 2022	Studies report at most small or fading changes in gaming and sleep; no robust health effect established. Registry value is the dated exit, not demonstrated success	Borrowed adult IDs and accounts; excluded games and devices; offline substitution
China: NPPA minor online-game notice; national regulation (National Press and Publication Administration (China), 2019)	Playtime and spending; under 18; $\leq$90 min/day (3 h holidays); no play 22:00 to 08:00; real-name verification	Eff. Nov 2019; time limits superseded by 2021 notice	Repeated-measures study: weekly play fell 3.5 h to 2.6 h in the high-risk group; no untreated counterfactual (Yang et al., 2023)	Adult-ID borrowing; family account sharing; account rental; unlicensed and offline games
China: NPPA strict limits; national regulation (National Press and Publication Administration (China), 2021)	Online play; under 18; 20:00 to 21:00 Fri/Sat/Sun and holidays only	Eff. Sep 2021	Same study as above; self-reports rather than platform logs; suggestive only (Yang et al., 2023)	ID and account markets persisted; firms added facial checks, raising rather than eliminating circumvention costs

Policy; instrument	**Object; ages; scope; stated mechanism**	**Dates; status (Aug. 2026)**	**Evaluation / measured first stage**	**Adaptation notes**
Japan: Kagawa Prefecture gaming ordinance (Kagawa Prefecture, 2020)	Gaming and smartphone time; minors; aspirational 60 min/day (90 holidays); evening phone cutoffs	Eff. Apr 2020; upheld by Takamatsu District Court 2022 in a challenge brought with a teenage plaintiff	No causal evaluation or measured first stage found	Non-penal "endeavor" duties; household noncompliance; enforcement rests on family mediation
France: Loi 2018-698 school phone prohibition (République française, 2018)	Device use; écoles and collèges (opt-in lycées); school premises and activities; pedagogical and disability exceptions	Eff. from the 2018/19 school year	No national causal evaluation with a measured first stage	Phones kept in bags; uneven confiscation; out-of-sight use; after-school displacement
France: Loi 2023-566 "majorité numérique" (République française, 2023)	Account opening; under 15 require parental authorization	Promulgated Jul 2023; application constrained pending implementing decree and EU-law coordination	None	False ages; adult-assisted signup; VPN and service substitution; unresolved age-assurance burden
France: "pause numérique" expansion (Ministère de l'Éducation nationale (France), 2025)	Full-day physical separation of phones; collège pupils (roughly ages 11 to 15); intensifies the 2018 law	Pilot in 2024/25; national expansion announced for 2025/26	No mature evaluation or first-stage estimate yet	Lockers vs. pouches vs. collection vary locally; secondary phones; staff burden; end-of-day bottlenecks
Netherlands: national classroom device guideline (Government of the Netherlands, 2024)	Phones, tablets, smartwatches in class; primary and secondary; educational and medical exceptions	Secondary Jan 2024; primary in 2024/25; sector agreement, not statute	Ministry-commissioned implementation research reported teacher-perceived concentration gains; observational, no common first stage	School-level variation (bags, lockers, collection); teacher-authorized use makes nominally identical schools different treatments
Brazil: Rio de Janeiro municipal decrees 53.019/2023 and 53.918/2024	Device use; municipal middle schools; classroom-only, then full school day; stored or silent; pedagogy, medical, emergency exceptions	Classroom decree Aug 2023; school-wide Feb 2024; later nested under federal law	Matched DiD across 325 schools: "most/all using" fell 52% to 19% vs. 40% to 15% in comparisons; scores +0.06 SD after the full-day phase; no significant classroom-only effect (Lichand et al., 2026)	72% of treated schools still reported daily phone-bringing; pockets and bags; bathroom access requests; container boxes adopted; principals reported implementation difficulty
Brazil: Lei 15.100/2025 national school-phone law (República Federativa do Brasil, 2025)	Device use; basic education nationwide; classes, recess, intervals; accessibility, health, emergency exceptions	Published Jan 2025; regulated during 2025	Early national surveys: about one-third of students unaware after five months; roughly one-quarter of schools storing phones, so legal reach exceeded realized treatment	Phones remained in pockets and backpacks; storage and enforcement capacity varies sharply

Policy; instrument	**Object; ages; scope; stated mechanism**	**Dates; status (Aug. 2026)**	**Evaluation / measured first stage**	**Adaptation notes**
Norway: primary-school generative-AI guidance (Norwegian Ministry of Education and Research, 2026)	Model use; grades 1 to 7 discouraged; teacher-supported introduction later; privacy and pedagogy rationale	Guidance 2026; soft law, not a statutory ban	No evaluation or behavioral first stage located	Unsanctioned tools at home or via personal accounts; municipal and teacher variation; substitution among models
UK: Online Safety Act children's duties and Ofcom Protection of Children Codes (Ofcom, 2025)	Platform systems, features, content; services likely accessed by under-18s; safety-by-design duties; age assurance where needed	Act Oct 2023; child-safety duties enforceable from Jul 2025	No mature causal evaluation; correct first stages are service design changes, age-assurance coverage, and exposure audits, not the enforcement date	Age misstatement and VPNs; migration to smaller or encrypted services; over-blocking; provider redesign
EU: Commission guidelines on protection of minors under DSA art. 28 (European Commission, 2025)	Platform design and features for minors; EU-wide; safer recommenders, age assurance, limits on addictive design	Guidelines Jul 2025; non-binding guidance interpreting a binding duty	None yet; prospective process outcomes via audits and enforcement cases	False ages; service migration; risk of exclusionary or over-collecting age assurance
US: Arkansas Social Media Safety Act (Act 689) (State of Arkansas, 2023)	Account opening; under 18; age verification plus parental consent	Enacted Apr 2023; enjoined 31 Aug 2023, the day before effect; permanently enjoined Mar 2024	None possible, since it never operated; coding Arkansas as treated after Sep 2023 is exposure misclassification	Litigation displaced implementation entirely
US: Utah Social Media Regulation Acts and 2024 replacement (State of Utah, 2023)	Account, features, use; under 18; parental tools, defaults, curfew provisions	Signed Mar 2023; postponed; replaced 2024; key provisions enjoined Sep 2024; litigation continues	None; the version churn itself demonstrates the need for provision-level status fields	Legislative replacement; platform litigation; likely false-age and VPN responses
US: Florida HB 3 (State of Florida, 2024)	Account; under 14 barred; ages 14 and 15 require parental consent	Signed Mar 2024; scheduled Jan 2025; core provisions preliminarily enjoined Jun 2025	None, as core enforcement was blocked; record injunction dates as separate registry events	Litigation; service reclassification; false ages; movement to excluded messaging and gaming services
US: Texas SCOPE Act (HB 18) (State of Texas, 2023)	Account, features, data practices; known minors; registration and parental tools	Signed Jun 2023; eff. Sep 2024; monitoring provisions enjoined Aug 2024; other duties operative	None; first stages must attach to surviving provisions, not the whole act	Provision-level injunctions altered the treatment bundle; provider redefinition of covered services

Policy; instrument	Object; ages; scope; stated mechanism	Dates; status (Aug. 2026)	Evaluation / measured first stage	Adaptation notes
US: NYC Public Schools ChatGPT block and reversal (New York City Public Schools, 2023)	Model access; district devices and networks	Blocked Jan 2023; reversed toward guided AI literacy May 2023	No causal evaluation or measured first stage found	Immediate substitution to personal devices, home networks, and other models; model-level blocks proved brittle and short-lived
US (California): SB 243 companion chatbots (California State Legislature, 2025)	Companion-chatbot features and use; all users with added minor protections; disclosures and crisis protocols	Enacted 2025; operative requirements from 2026	None yet. Prospective metrics: verified under-18 access, crisis-protocol quality, sexual-content exposure, session length, dependency cues, provider exits	Age misstatement; open-source or foreign services; euphemistic prompting; relabeling away from "companion" functionality
US: GUARD Act (S. 3062), federal proposal (United States Congress, 2025)	AI-companion access; under 18; proposed age verification, companion ban, chatbot-status disclosures, and limits on sexual and self-harm-related interactions	Introduced Oct. 2025; reported with a substitute from Senate Judiciary and placed on the Senate calendar May 2026; not enacted as of Aug. 2026	No outcome evidence; treat legislative progress only as evidence of policy attention	Possible evasion through narrow product definitions, along with constitutional and privacy challenges
US/global: FTC companion-chatbot inquiry and Character.AI policy changes (Federal Trade Commission, 2025; Character.AI, 2025)	Account, model, and feature restrictions affecting minors; provider disclosures, crisis protocols, and under-18 access restrictions	FTC orders issued Sep. 2025; Character.AI changes announced in 2025 and rolled out across 2025 and 2026	No independent outcome evaluation. Needed: verified under-18 reach, bypass rates, exposure to attachment, self-harm, and sexual dialogues, parental-control uptake, and cross-service substitution	False birth dates, adult accounts, open models, displacement from monitored platforms, and provider-policy churn should be measured and archived

A.2. Question bank for policy and research, by TRACE lens

TRACE lens	Questions for policy and research
Treatment	What object, use, age, time, and place are restricted? At which governance level is the decision made and enforced? What exceptions and approved alternatives exist? Which causal mechanism connects this design, rather than technology exposure in general, to the stated goal?
Reach	Who knows the rule, can comply, and encounters enforcement? Does observed use or access actually change, by how much, and for whom? What do compliance, age-assurance errors, teacher discretion, and enforcement labor cost? Can the first stage be measured without monitoring children's private communications?

TRACE lens	**Questions for policy and research**
Adaptation	Which services, devices, accounts, models, locations, or times replace the restricted activity? Does displacement preserve the hypothesized harm or create a different one? Does use become less visible to parents, educators, or safety systems? How do responses evolve as children, schools, and platforms learn?
Consequences	Which proximal mechanism should move first, and on what timeline? What is the credible counterfactual? Are effects sustained after novelty and initial enforcement shocks? What are the spillovers to home, peers, teachers, and non-targeted platforms? What evidence would falsify the policy's theory of change?
Equity/equity/exp	Who receives benefits and who bears privacy, disciplinary, accessibility, or opportunity costs? Are accommodations usable without stigma? How do children define safety and acceptable trade-offs? Which subgroup harms or null effects trigger revision? When should the policy be retained, narrowed, replaced, or sunset?